Hie-VLA: A Hierarchical Vision-Language-Action Model for Learning Long-horizon Robotic Manipulation from Atomic Skill Dataset

***Abstract*— Balancing task analysis and execution efficiency is a critical challenge in robotic manipulation. While existing Vision-Language-Action (VLA) foundation policies have shown that using pre-trained vision-language models (VLMs) as the backbone enhances continuous action generation, they still struggle with reasoning subgoals when executing coherent, sequential subtasks in long-horizon tasks. To address this issue, we propose decomposing tasks from the perspective of skill primitives by constructing an atomic skill dataset that facilitates implicit learning of multi-step action capabilities. We present a hierarchical training policy framework that aligns subtask instructions with corresponding atomic skills, coordinating reasoning and execution components to enhance the model's ability to understand observations in dynamic environments and execute subtask actions effectively. Finally, extensive experiments on real robots demonstrate that our model outperforms previous state-of-the-art methods by 11% in long-horizon tasks and demonstrates reliable task reasoning in unseen scenarios.**

I. INTRODUCTION

Vision-Language-Action (VLA) models represent a promising paradigm [1], [2], [3], [4], [5]. By leveraging pretrained vision-language models (VLM), they map multi-modal inputs (e.g., text and image) into a unified semantic symbolic space, and associate them with continuous action features to generate practical control policies for diverse daily manipulation tasks. However, it faces challenges in generalizing to novel scenes or unfamiliar objects. When encountering new tasks, it exhibits reactive behaviors driven by prior training, akin to muscle memory, rather than performing perceptual reasoning before action execution in unseen situations. This limitation significantly hinders the deployment of robots, especially in *long-horizon* tasks that require multi-step reasoning and diverse interaction.

To address this, prior research has established reliable frameworks that rely on large open-source VLM to generate chain-of-thought (CoT) reasoning sequences from observation for task decomposition [6], [7]. These frameworks embed VLM thinking into visual representations, assisting VLA during inference. But most of these systems lack understanding of the action space, as they do not truly understand the underlying skills required for task execution, and merely rely on model judgments during reasoning to guide execution. In contrast, when humans are asked to organize objects, they first identify object types, then determine appropriate skills for execution. This process enables them to both recognize target objects and decompose the scene into subtask movements through skill composition. Inspired by this, explicit skill categorization strengthens visual understanding in complex environments.

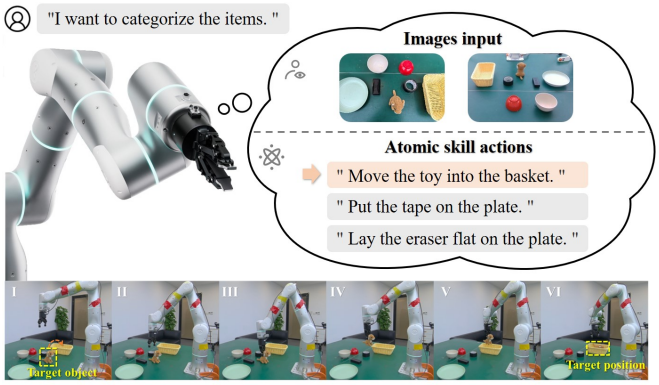

Fig. 1: We propose **Hie-VLA**, a manipulation model that bridges the gap between vision-language reasoning and robotic skill execution for long-horizon tasks, enabling closed-loop control during asynchronous inference.

To realize this potential, we hypothesize that training VLMs on datasets with skill annotations and structured language representations enables them to implicitly learn action semantics grounded in instruction prompts. However, most embodied datasets [8], [9] lack intermediate subtasks; a natural solution is to break down end-to-end tasks into sequential steps and annotate each step manually. A key difficulty is that this approach is susceptible to human bias, as the same task may be decomposed in multiple ways. This necessitates a more systematic approach to ensure consistency in training dataset construction.

To bridge this gap, we propose merging discrete primitive data, which we name atomic skills, to construct a new training dataset paradigm. Each atomic skill represents a single real-world action (e.g., pick, grasp, or place). While previous work [10] found that atomic datasets effectively learn individual skills, complex real-world tasks require executing multiple primitives across several steps. Therefore, we leverage large language models (LLM) to compose atomic skills into extended sequences, generating scalable synthetic datasets for long-horizon tasks. Furthermore, we employ **Hie-VLA**, a hierarchical framework that connects this data across abstraction levels, integrating high-level reasoning with low-level execution.

In contrast to end-to-end approaches, Hie-VLA requires two-stage training: it first infers linguistic subtasks aligned with atomic skills, then predicts actions based on them. The intuition behind our design is the integration of *'planning' and 'execution'* capabilities. For instance, consider a robot

tasked with tidying a desk: instead of rigidly executing a predefined fixed sequence, the robot should understand human instructions such as "I want to categorize the items," respond to subtask prompts such as "Move the toy into the basket," and predict suitable interactive actions, as illustrated in Fig. 1. We validate the proposed model through extensive experiments on the Flexiv robot, and the results reveal that our approach achieves superior performance over state-of-the-art VLAs.

The contributions of this work are summarized below:

- We propose a new training paradigm that implicitly learns long-horizon manipulation through datasets constructed from atomic skills.
- We present a hierarchical framework that bridges task reasoning and skill primitives through linguistic subtask inference, making task execution interpretable.
- We apply an asynchronous inference strategy to achieve closed-loop inference. Extensive experiments show that our model can generate reliable subtask instructions and effectively generalize to novel situations..

II. Related Works

A. Dual-System VLA for Long-Horizon Manipulation

Recent Vision-Language-Action (VLA) models integrate vision-language models (VLMs) and action controllers through end-to-end training, showing substantial promise for sophisticated manipulation tasks [2], [3], [11], [12], [13]. However, these methods suffer from limited action generation frequency and inadequate exploitation of VLM reasoning. Considering these shortcomings, hierarchical planning [14], [4] enhances reasoning by decomposing tasks into simpler subtasks that can be executed directly. Building upon this, GR00T-N1 [15] proposes a dual-system architecture where VLMs extract latent features as conditions for diffusion-based action generation. Furthermore, Fis [16] introduces a uniform inference strategy to accelerate the two-stage framework. These methods rely on a hierarchical processing approach to strengthen correlations between components, but they struggle to generate consistent multi-step actions due to the segmented nature of subtasks. In contrast, our work addresses this limitation by learning skill-based subtask concatenation, enabling fluent execution of long-horizon tasks.

B. Reasoning Guidance for Action Generation

Early approaches show that employing large language models (LLMs) for step-by-step inference can effectively construct chain-of-thought (CoT) reasoning for task comprehension [17], [18], [19]. Several works have explored similar techniques for high-level reasoning in robotic task planning, where abstract subgoals are translated into coordinated action policies. Additionally, ECoT [7] adopt visual observation to provide perceptual information for task decomposition. CCoT [20] utilizes object attributes for task classification. These methods rely on pretrained VLM capabilities and manual rules for task decomposition, which has become standard practice in VLA model training. However, we argue that (1) rule-based decomposition relying solely on LLM reasoning struggles with dynamic environments and unexpected events (e.g., object drops), and (2) VLMs exhibit perceptual biases and excessive inference latency, making it necessary to ground high-level task decisions in reliable primitive actions. In contrast to prior works, our approach trains a VLA policy that decomposes complex long-horizon tasks into atomic skills by aligning subtask instructions with skill trajectories, enabling robots to learn corresponding execution behaviors.

C. Data for Policy Scaling

Embodied manipulation aims to train policies that generalize across diverse scenarios. Towards this goal, recent works enhance generalist policies by collecting diverse robot datasets [8], [9], [21] or generating simulation data [22]. MimicGen [23] generates large-scale data by segmenting demonstrations into object-centric subtasks and applying spatial transformations. DexMimicGen [24] extends this to humanoid multi-arm coordination, while [25] reduces data requirements by synthesizing training data from single demonstrations via 3D point clouds. Other works leverage human videos [26] and non-robot data [4] to enhance understanding of environments and actions. Another line of research [10] demonstrates that skill libraries constructed via task decomposition enable data-efficient embodied manipulation. While these methods lack effective scaling to multi-step tasks, we learn long-horizon skills from synthetically composed atomic skill data using pre-trained vision-language model. To our knowledge, this is the first VLA trained on concatenated atomic skill datasets for long-horizon tasks.

III. METHODOLOGY

Problem Formulation. VLA models typically learn robotic control policies from multi-modal inputs via imitation learning, thereby enhancing environmental plasticity and task adaptability. Formally, our goal is to improve the ability of VLA models in multi-stage planning and reasoning chain construction for long-horizon complex task.

Overview. The core idea of our approach is twofold: encourage the model to **(A)** deliberate on the atomic skills required for the task and determine the next subtask to execute, and **(B)** increasingly ground each step of the reasoning chain in the visual observation and robot state before action prediction.

We visualize the model architecture that we train the VLA to perform for an example task in Fig. 2. In this work, we construct comprehensive observations, including the robot state a_t, multi-view images o_t, high-level prompt l_h, and subtask instruction l_s. We train model π conditioned on these observations to maximize the likelihood of generating temporally extended action sequences $a_{t:t+H}$, this corresponds to the following optimization problem:

$$\max_{\theta} \mathbb{E}_{(a_{t:t+H}, o_t, l_h, l_s)\sim\mathcal{D}} \left[\log \pi_\theta\big(a_{t:t+H} \mid o_t, l_h, l_s\big)\right] \quad (1)$$

where D represents demonstration datasets in model training, and a_t denotes 7 degrees of freedom (DoF) pose control,

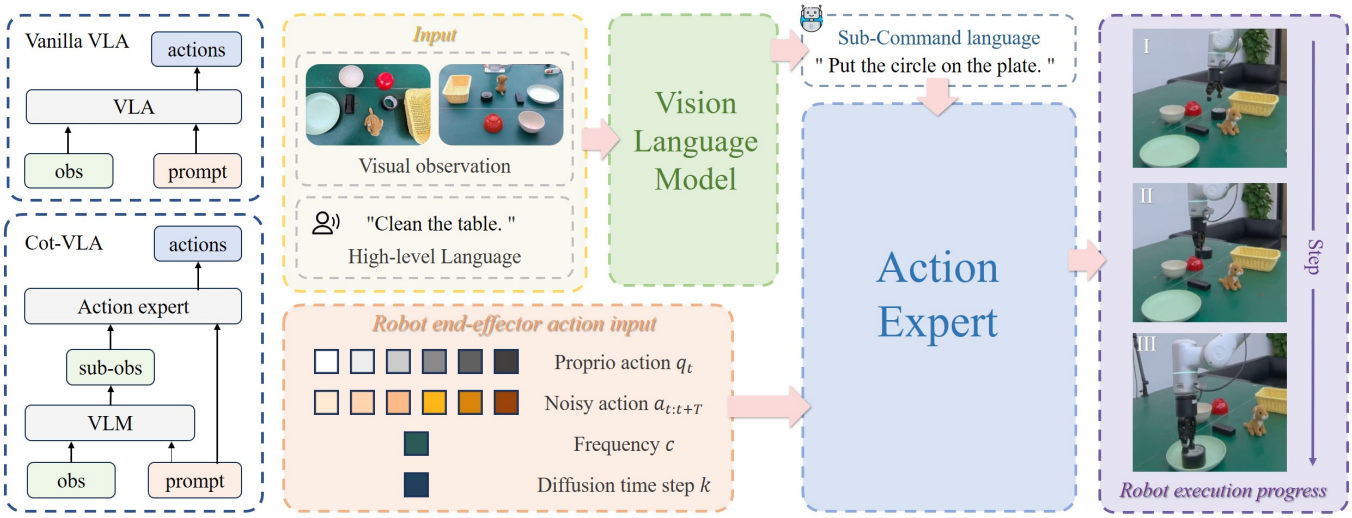

Fig. 2: **Overview of our architecture.** *Left top*: Vanilla VLA directly maps high-level goals and observations to actions. *Left bottom*: The planner predicts subgoal images, which guide the controller to generate corresponding trajectories. *Right*: Hie-VLA employs a hierarchical architecture that integrates high-level task planning with low-level motion control. The planner decomposes tasks into sub-atomic skills, while the controller generates executable actions via flow matching.

consisting of 3-DoF absolute position and 3-DoF rotation of the end effector along with 1-DoF gripper state. During the execution of the inference, the model reasons about subtask instructions based on current observations to guide robot atomic task execution. We describe each part in detail below.

A. Model Architecture

Vision Encoder. We employ SigLIP [27] to extract visual representations that capture spatial details. Specifically, the model concatenates the exterior and wrist images, with all inputs resized to 224×224 pixels. These combined inputs are projected into a shared embedding space for semantic alignment.

LLM backbone. The Gemma-2B is adopted as the LLM backbone for Hie-VLA. [4] demonstrates its capability for reliably decomposing complex instructions. Therefore, we repurpose the high-level instruction decomposition model to condition on low-level atomic skill actions, coupling them with the decomposed subtask instructions.

Hierarchical control mechanism. In VLA models, the policy receives only the final goal and lacks access to subtask information. Consequently, it must rely on implicit or explicit planning to infer intermediate steps for task execution. The vanilla VLA [1] (Fig. 2, left top) is limited to generating actions based on high-level prompts, without the ability to produce language outputs. CoT-VLA [28] (Fig. 2, left bottom) introduces a new training paradigm to predict intermediate images for action guidance, but struggles to classify them into specific skills.

To address this, we employ a hierarchical control strategy to leverage the instructional information processed by the model. The VLM serves as a high-level planner, reasoning about the next atomic subtask, and the low-level controller generates actions to execute it. Equation (1) can be formulated as follows:

$$\pi_\theta^{\text{planner}}(o_t, l_h) \rightarrow l_s, \quad \pi_\theta^{\text{controller}}(o_t, l_h, l_s) \rightarrow a_t \qquad (2)$$

B. Atomic Skill Dataset for Long-Horizon Tasks

The training of Hie-VLA relies on a collection of demonstrations (l_h, γ_t) with $\gamma_t = (o_t, l_s, a_t)$. The primary challenge lies in obtaining subtask annotations for real-world long-horizon tasks at scale, which necessitates extensive human intervention. Previous works used manually designed rules to decompose tasks into atomic skill libraries. However, such top-down decomposition into fixed subtasks creates rigid structures prone to error accumulation and limits generalization. Therefore, we construct the dataset at the atomic skill level, composing diverse long-horizon tasks through flexible skill combinations, as illustrated in Fig. 3.

Atomic Skill Collection. We define each atomic skill as an independent operational task, where the robot interacts with objects to accomplish specified tasks (e.g., pick, move, grasp). Grasping different objects within the same scene can be treated as multiple independent atomic operations. Each demonstration is annotated with a descriptive prompt (e.g., "Put the eraser on the plate"). These annotations serve as the subtask prompts l_s in our framework. At the end of each atomic trajectory, the robot returns to the preset starting pose to reduce motion discontinuity between skill transitions.

Atomic Skill Remapping. For simplicity, we assume each atomic skill can be completed within a single time step. Given a trajectory of atomic skills $(\gamma_1, \gamma_2, \ldots, \gamma_n)$, the semantic abstraction module leverages LLMs (e.g., GPT-4o) to generate a coherent high-level prompt l_h (e.g., "Clean the table") by combining these discrete atomic skills into a unified abstract instruction. Specifically, the module takes the annotations of the selected atomic skills as input, samples and concatenates their corresponding action sequences, and

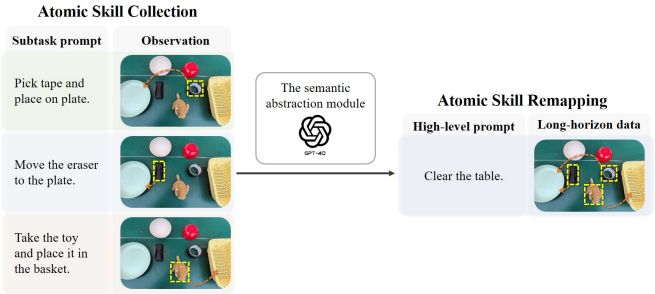

Fig. 3: **The process of constructing Atomic skill data**. Taking the Desktop Organization task as an example, three discrete independent object grasping skills are merged into a temporally continuous long-horizon task with high-level semantics.

synthesizes temporally extended trajectories that align with the abstracted high-level prompt. (e.g., "Pick up the circle and place it on the green plate" → "Grasp the toy and place it in the yellow basket" → etc.). These synthesized trajectories form our training dataset, which is created through two data merging methods: **(a)** randomly remapping, where for multiple atomic skills associated with the same task, we randomly vary the composition order and the number of skills combined (e.g., single skill instances), and **(b)** sequential remapping, where we serially chain multiple action skills in chronological order. Our goal is to enhance data diversity while preserving inherent reasoning relationships between atomic skills, ensuring that high-level instruction l_h maintains logical consistency for long-horizon tasks.

C. Training Paradigm

To enable continuous reasoning across atomic skills and prediction of high-frequency action within each skill, the model training is divided into two stages. The first stage aims to enhance the VLM's ability to identify atomic skills within the environment, and the second stage specializes the model for skill control as well as efficient inference.

The first training stage optimizes the language model backbone while freezing the image encoder and linear projection layer. As shown in [4], VLM training with cross-entropy loss between text tokens and predicted logits can achieve excellent language following performance. However, this fails to align with the atomic action space and may lead to incorrect action guidance during inference. Therefore, we adopt a tokenizer [29] to compress action chunks into discrete tokens a_s (i.e., actions as a *special language*), concatenate them with the subtask prompt l_s into a unified token sequence, and train the model with separate losses for subtask and action prediction through autoregressive language modeling. The training loss is defined as:

$$\mathcal{L}_s = \mathbb{E}_{\mathcal{D}}\left[H(l_s, f_\theta^s(\mathbf{o}_t, l_h)) + \lambda \cdot H(\mathbf{a}_s, f_\theta^a(\mathbf{o}_t, l_s))\right] \quad (3)$$

where $H(l_s, f_\theta^s(\mathbf{o}_t, l_h))$ and $H(\mathbf{a}_s, f_\theta^a(\mathbf{o}_t, l_s))$ are the cross-entropy losses for text tokens and discrete action tokens, respectively, and λ is a tunable hyperparameter that balances their weights for better high-level instruction grounding.

After training the VLM with discrete tokens for 100k gradient steps, we freeze it and perform a second stage to train the action expert. Following [30], we train the action expert *without propagating gradients back to the VLM*. This preserves semantic knowledge during training. We use flow matching policy to generate actions matching high-frequency robot motion in the real world. At timestep $\tau \in [0, 1]$, the noised action chunk is computed as $A_t^\tau = \tau A_t + (1-\tau)\epsilon$, where $A_t = [a_t, a_{t+1}, \ldots, a_{t+H-1}]$ denotes the ground-truth action chunk and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is Gaussian noise. The model $f_\theta(\phi_t, A_t^\tau, q_t)$ learns the denoising vector field $\epsilon - A_t$ by minimizing:

$$\mathcal{L}_f = \mathbb{E}_\tau\left[\|f_\theta(\phi_t, A_t^\tau, q_t) - (\epsilon - A_t)\|^2\right]. \quad (4)$$

where q_t represents the robot state embeddings and ϕ_t indicates vision-language embeddings from the VLM

D. Inference Process

Inference Strategy. Inference speed is a key challenge for VLA policies, because the VLM reasoning time does not align with action control frequency. There are two strategies for accelerating policy inference: (1) *synchronous execution*, where the model performs high-level reasoning every N steps, and (2) *asynchronous execution*, in which the high-level planner continually updates skill prompts, while the low-level controller receives subtasks after K steps to predict the next action chunk. Asynchronous inference fixes the reasoning chain for multiple steps, making subtask prediction more frequent than action generation, creating a closed-loop feedback mechanism between execution and reasoning. We compare the trade-off between performance and inference speed in Table I.

Temporal Action Fusion. During inference, action distributions vary across atomic skills. This results in visible pauses between action chunks that not only cause execution latency but also disrupt robot dynamics. To alleviate these, we employ an adaptive action ensemble strategy [31], which fuses the current action with historical actions to reduce discontinuities in the execution trajectory.

In addition, atomic skill prompts can be generated via *VLM environmental reasoning* or *user interaction* to execute specified actions, with explicitly defined tasks taking highest priority. The policy execution supports real-time user intervention to update information, provide feedback, or completely change the task. This is a practical method to test the instruction-following ability in Human-Robot Interaction (HRI). For example, during the *Desktop Organization* task, if the user instructs to grasp the toy, the model should prioritize executing that atomic skill action (as shown in Fig. 6).

IV. Experiments

We aim to address the following research questions: **(1)** Can the model handle long-horizon tasks and generalize

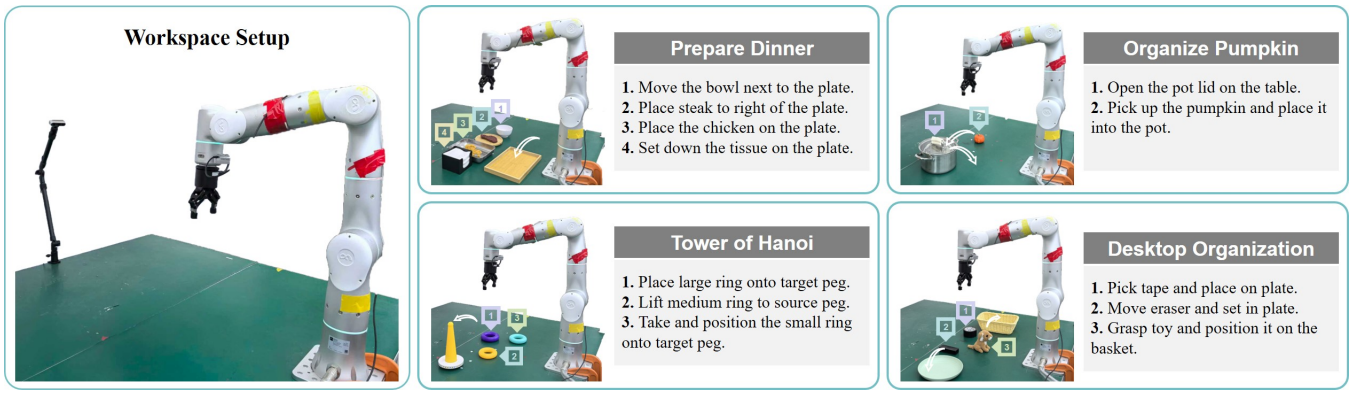

Fig. 4: **Robot setup and task visualization for real-world manipulation tasks.** We evaluate Hie-VLA with four tasks and each task consists of multiple atomic skills.

TABLE I: Real-World Evaluation of Hie-VLA and Baselines Across Multiple Tasks. All Methods are Trained Using The Same Atomic Skill Dataset and Success Rates are Reported.

Method	Organize Pumpkin		Desktop Organization			Tower of Hanoi			Prepare Dinner				Task(all)
	1	2	1	2	3	1	2	3	1	2	3	4	Avg.
RDT	71.4	57.1	62.5	40.2	67.5	44.3	22.5	10.3	78.3	54.2	68.4	27.2	50.3
π_0	72.6	45.4	87.5	54.5	69.2	38.2	18.5	8.6	84.2	62.4	71.5	40.6	54.4
$\pi_{0.5}$	78.5	67.5	84.5	67.5	72.7	81.5	54.6	20.5	85.0	78.5	76.0	45.5	67.7
Ours	**86.5**	**77.5**	**92.5**	**78.2**	**80.8**	**89.8**	**67.7**	**49.0**	**89.6**	74.7	**78.5**	**50.0**	**76.2**

TABLE II: Real-world Generalization Evaluation of The Flexiv Robot on Unseen Objects, Backgrounds and Lighting.

Method	unseen obj	unseen lighting	unseen bg	Avg.
RDT	57.5	22.5	18.1	32.7
π_0	62.5	54.2	55.8	57.5
$\pi_{0.5}$	65.0	62.5	52.5	60.0
Ours	**72.5**	**78.2**	**75.0**	**75.2**

to unseen situations (Section IV-B)? **(2)** How do discrete action tokens and atomic skill data composition influence model performance (Section IV-C)? **(3)** What is the impact of different inference strategies on model performance (Section IV-D)? **(4)** We demonstrate the effectiveness of atomic skill datasets for multi-step tasks (Section IV-E).

A. Experimental Setup

Teleoperation Device. To collect the end-effector pose, we use Noitom Robotics, which is a portable wearable device for teleoperation. It captures human hand motion and generates corresponding robot actions through retargeting. This device provides high-fidelity motion capture for intuitive control and high-precision data acquisition.

Robot Configuration and Tasks. The experimental setup is shown in Fig. 4. We employ a Flexiv Rizon 4 robot with a standard parallel gripper as the end-effector. The system employs two Intel RealSense D435 cameras to capture visual observations, one providing a third-person view and the other mounted on the wrist for egocentric perception. We develop a suite of real-world tasks to evaluate the efficacy of our proposed method, as listed below:

- *Organize Pumpkin*: This task consists of two atomic skills: (1) open the pot lid on the table, and (2) pick up the randomly placed pumpkin and put it in the pot.
- *Desktop Organization*: Three objects (i.e., tape, eraser, toy) need to be placed in their designated positions (i.e., plate, basket), requiring three atomic pick-and-place skills.
- *Tower of Hanoi*: Move the rings from the source peg to the target peg following the rules, where ring size $\in \{small, medium, large\}$. Only one ring can be moved at a time, and a larger ring cannot be placed on top of a smaller ring.
- *Prepare Dinner*: Complete the task through a four-step sequence: moving the white bowl, placing the steak on the dinner plate, placing the chicken beside the steak, and setting the napkin. All items must be arranged in the correct spatial configuration.

Implementation Details. The training process consists of two stages: in the first stage, we train the VLM by uniformly sampling from all atomic skill datasets; in the second stage, we fine-tune the model on task-specific atomic skill datasets. We train the model using the AdamW optimizer with weight decay and gradient clipping, leveraging Fully Sharded Data Parallel (FSDP) for efficient distributed training. In stage one, we train with 8 NVIDIA A100 GPUs (80GB VRAM), using a per-device batch size of 32 and a learning rate of 5e-5. In stage two, we reduce the learning rate to 1e-5 while keeping other settings unchanged.

B. Real-world Evaluation

Baselines. To comprehensively evaluate our model, We

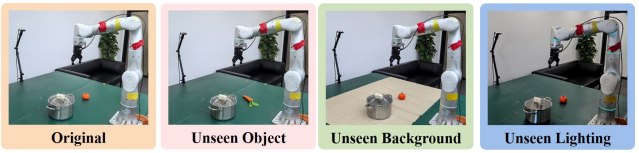

Fig. 5: Comparison of the performance across different accuracy in unseen scenarios.

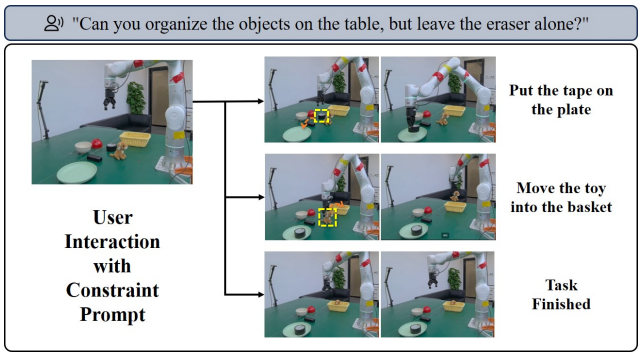

Fig. 6: **The process of executing actions under language constraints**. The robot avoids restricted skills based on constraints specified in the dialogue.

employ strong baselines in robotic manipulation: RDT, π_0, $\pi_{0.5}$. We finetune all models on our collected atomic dataset, initialized from publicly available pretrained weights, under the same experimental setup. Since baseline models lack subtask annotation input, we *only use high-level instruction* as language prompt.

Training Data. We collected trajectory data samples for finetuning the backbone VLA model, with the number of demonstrations for each task detailed below:

- Desktop Organization: We collect 30 demonstrations for each atomic skill and *randomly remap* them into 150 training samples.
- Prepare Dinner: We collect 40 demonstrations per atomic skill and *randomly compose* them into 250 training samples.
- Organize Pumpkin: We collect 30 demonstrations for each atomic skill and concatenate them according to the fixed sequence: *lift up pot lid* $\rightarrow$ *place the object*.
- Tower of Hanoi:We collect 20 demonstrations for each atomic skill and concatenate them following the fixed order of placing rings from *largest to smallest*.

Metric. Our main evaluation metric is the task success rate. Task evaluation proceeds as follows: skills without sequential dependencies are evaluated independently; otherwise, they are tested in order, with any unsuccessful action leading to immediate termination. Each task is evaluated 300 times, with success requiring all atomic skills to succeed.

Result. To ensure a fair experimental evaluation, all baseline models are finetuned on the same atomic skill demonstrations. We present the results in Table I, which show that our model improves the success rate of each atomic skill in long-horizon tasks, achieving an average success rate of 76.2% across 4 tasks and surpassing the previous SOTA methods RDT and $\pi_{0.5}$ by margins of 33% and 11%, respectively. Through embedding subtask labels in VLM training, our model leverages pretrained knowledge for task decomposition and guided action generation. This approach enables Hie-VLA to achieve superior performance on 11 out of 12 atomic tasks, demonstrating the robustness of its skill execution.

TABLE III: Success Rates of Instruction Following across Different Constrained Unseen Instructions

Unseen Instruction	Following Rate(%)
Only open the pot lid on the table.	77.5
Organize the table except the eraser.	82.8
Prepare a dinner meal excluding chicken.	65.0

TABLE IV: Performance of the Desktop Organization Task under Different λ Values

Task	No.	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 2$	$\lambda = 5$
Desktop Organization	1	87.2	**92.5**	85.0	76.5
	2	76.5	**78.2**	72.5	62.5
	3	78.5	**80.8**	76.5	70.0

Generalization Experiment. To evaluate the generalization capability of Hie-VLA in real-world settings, we conduct experiments on the "Organize Pumpkin" task across three Out-of-Distribution (OOD) scenarios: unseen manipulated objects, low lighting, and complex backgrounds, as illustrated in Fig. 5. These scenarios assess whether the model can extract task-relevant information while remaining robust to distractions. Table II shows that our model demonstrates strong generalization capabilities and outperforms other baselines. These results highlight the importance of staged training with heterogeneous modality inputs in Hie-VLA systems to improve robustness to perceptual perturbations.

Fig. 6 shows that under constrained prompts, the model executes specific actions while avoiding picking up the eraser. We systematically assess performance across multiple tasks with unseen prompts in Table III, which shows that our model has better language following capabilities. There are two reasons: **(1)** Hie-VLA can understand instruction constraints and provide correct subtask prompts for action generation. **(2)** The atomic skill dataset enriches data diversity by incorporating high-level semantic information across different action combinations.

C. Ablation Study

Discrete Actions Inference. We trained Hie-VLA with and without the tokenizer to generate discrete actions that align with subtask annotations during VLM training and evaluated their success rates on the Hanoi task. Fig. 7 presents the results, which clearly illustrate that adding action information enhances subtask success rates. It indicates that

TABLE V: Performance of the Prepare Dinner Task under Different Dataset Compositions

Task	No.	randomly	sequential	mixed
Prepare Dinner	1	78.5	86.4	**89.6**
	2	69.8	61.2	**74.7**
	3	72.5	65.4	**78.5**
	4	54.2	45.6	50.0

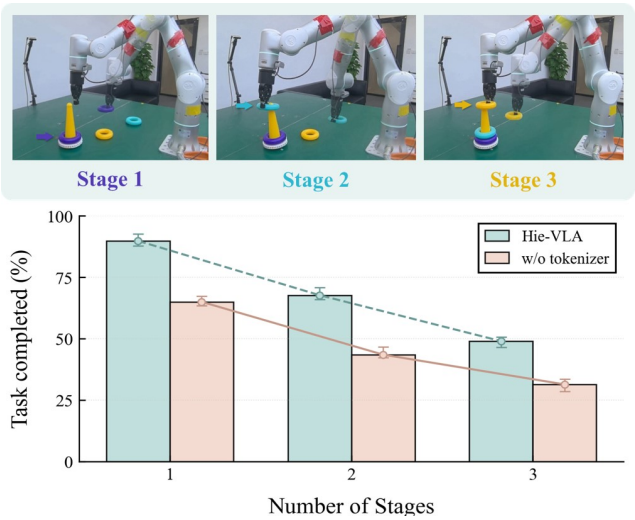

Fig. 7: Performance comparison between Hie-VLA and w/o-tokenizer across different stages in the Tower of Hanoi task.

discrete actions facilitate the distinction between different atomic skills and reasoning about correct skill instructions in dynamic environments.

Analysis of Loss Weights. Here we analyze the hyperparameter sensitivity of the model optimization loss function in (3). In Table IV, we examine the impact of λ with values set to 0.5, 1, 2, and 5. The experimental results demonstrate that the overall task success rate is optimal when $\lambda = 1$. As λ increases, the overall success rate decreases significantly, with actions becoming inconsistent with atomic skill instructions, suggesting that unbalanced loss weighting hinders joint optimization of language understanding and action execution.

Analysis of Dataset Composition. Next, we analyze the influence of atomic skill dataset composition. We surprisingly find that mixing randomly and sequentially remapped datasets improves subtask success rates, as shown in Table V. We attribute these gains to expanded training data from diversified compositional sampling, which improves the generalization and robustness of atomic skill execution. Notably, for sequential tasks, mixing datasets severely affects execution order and overall performance.

D. Efficient Inference Strategies for Model Execution

In Section III-D, we present different inference strategies for our model. We evaluate performance and inference time on the *Organize Pumpkin* task, as shown in Table VI. We use

TABLE VI: Influence of Different Inference Strategies on Model Performance. Avg Time Refers to the Average Inference Time Required to Complete the Task.

Method	Success Rate(%)	Avg Time(s)
Naïve	76%	55.4
Synchronous	73%	42.5
Asynchronous	80%	30,8

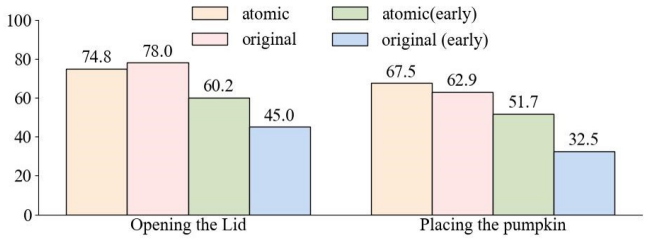

Fig. 8: Performance comparison of models trained on the atomic and original datasets across two sequential stages of the Organize Pumpkin task.

N=1 as the naïvely baseline, and N=5, K=15 for synchronous and asynchronous strategies respectively, as they provide the best performance-speed tradeoff. It can be observed that compared to the naïvely baseline, all strategies achieve faster inference, with asynchronous execution delivering the best performance and a 29% reduction in inference time. However, it doubles the compute cost as both the high-level planner and low-level controller execute in parallel.

E. Benefits of Atomic Skill Dataset

To evaluate atomic skill dataset optimization, we collect an equal amount of long-horizon data (two-step sequences) for the *Organize Pumpkin* task as the baseline, denoted the original dataset. We finetune $\pi_{0.5}$ on each dataset for 140K steps using 8 RTX 4090 GPUs, with results compared in Fig, 8. This shows that atomic training achieves comparable success rates while better learning sequential dependencies. Moreover, atomic training achieves higher success rates in half the training steps (denoted as **early training**), as the atomic dataset provides richer action diversity through skill combinations, improving data efficiency.

Notably, atomic skill collection is substantially faster than recording full multi-step trajectories. This provides a *scalable approach* for long-horizon task datasets, where diverse atomic skills are compositionally synthesized into varied trajectories.

V. CONCLUSIONS

In this paper, we introduce Hie-VLA, a novel hierarchical VLA foundation model that addresses the challenge of learning coherent multi-step behaviors. We demonstrate that hierarchical VLA can achieve long-horizon capabilities through compositional skill training, transforming environmental understanding into executable atomic skills and implicitly

learning coherence across multiple subtasks. Extensive real-world experiments show that our approach outperforms existing methods and generalizes to unseen environments and novel instructions.

Limitations. Despite these promising results, our approach has several limitations. The model may fail to distinguish target objects from visually similar items due to VLM hallucination during subtask reasoning, leading to incorrect skill execution. We hypothesize that scaling the atomic dataset to large-scale benchmarks (e.g., OXE dataset [8]) will both improve robustness to perceptual ambiguities and unlock the full potential of atomic skills for generalizable manipulation, which will be a key focus of our future work.

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